FMSG: Cyber: Cloud Molecular Beam Epitaxy

1 Enabling Future Manufacturing

Background and major challenges Thin-film material manufacturing has been a fundamental driving force behind new scientific discoveries and technological innovations in modern society. Advanced semiconductor thin films have enabled cutting-edge information technologies, while low-dimensional quantum materials paved the road towards a quantum revolution in which information will be processed in disruptively new modalities. The U.S. National Science Foundation (NSF) has identified the manufacturing of advanced materials as a top priority area, emphasizing the need for the U.S. to reestablish its leadership after two decades of investment decline [1]. At this critical juncture, traditional methods of thin-film material manufacturing, which rely on laborious human-driven processes, may struggle to keep pace with the growing demand for rapid creation, characterization, and optimization of materials. Moreover, materials synthesis, characterization, and modeling need to be strongly connected in a closed loop. Currently, these critical elements are often manually controlled in separate laboratories and facilities. This human-led integration of materials synthesis and characterization has led to conflicting interpretations, misinformed decision-making, slower convergence of material knowledge, and significant overhead in training the next generation of American material manufacturing workforce.

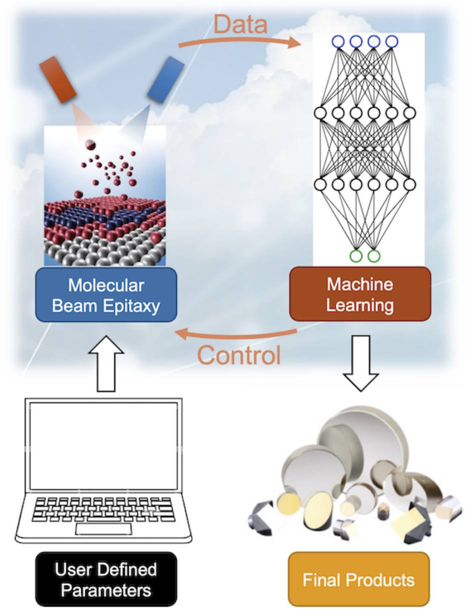


Figure 1: **Scheme of Cloud Molecular Beam Epitaxy.** User-defined specifications are sent to an artificial-intelligence-informed molecular beam epitaxy system. The characterization-control loop leads to a fully automatic convergence of the growth process.

Transformative impact of the proposed research With the recent rapid development of artificial intelligence (AI) and robotic material processing [2], it has now become feasible to envisage a completely new scheme where thin-film manufacturing and characterization can be automated and guided by advanced machine learning algorithms (see Fig. 1). The experimental data, such as synthesis conditions and characterization results from *in situ* transport, spectroscopy, and microscopy measurements, provide opportunities to train these algorithms, which in turn will guide the next iteration of material manufacturing. Once fully developed, this closed-loop system will have the capacity to operate autonomously, continuously, and efficiently, fabricating materials and devices with minimal human intervention. Ultimately, both the automated experimentation and data infrastructure could be moved to the cloud, benefiting not only the specialized material experts, but also the wider academic community and industrial partners that utilize advanced materials.

Vision statement **Here we aim at establishing a prototype system for the concept of Cloud Molecular Beam Epitaxy (*Cloud MBE*), which integrates state-of-the-art thin film manufacturing with artificial intelligence modeling to create a fully autonomous, self-learning robotic platform for producing quantum materials according to user-defined specifications.**

MBE is a thin-film deposition technique that involves precisely spray-printing dilute molecular beams onto substrates [3]. Traditionally, MBE operators search exhaustively in high-dimensional space defined by growth parameters like source fluxes, flux ratios, and substrate temperatures. In this project, the deposited thin films will be primarily characterized through electron diffraction patterns. This characterization data will be used to train a machine learning model, which will then predict the necessary adjustments to the growth parameters for the subsequent deposition iteration. This characterization-control cycle will function as a cloud-based manufacturing system, enabling the fabrication of materials and devices that meet user-defined specifications. To demonstrate this concept, we will assemble a cross-disciplinary team composed of a thin-film material scientist (**Yang**) and a computer scientist (**Chen**) who will collaborate on the production of superconducting materials such as FeSe and aluminum, which are poised to play a crucial role in the emerging field of quantum technologies.

Global context of the proposed work The U.S. is no longer the dominant player in the world in advanced thin-film deposition. The industrial labs that sponsored the great materials revolutions of the last century, such as Bell Labs and IBM Research, are no longer supporting materials discovery and crystal growth. Recent major discoveries in thin-film materials, such as the Quantum Spin Hall Effect [4] and the Quantum Anomalous Hall Effect [5], were made in Germany and in China, respectively. The Japanese materials science community has pioneered in machine-learning (ML)-informed MBE synthesis on $SrRuO_3$ [6, 7, 8, 9] and TiN [10], as well as the classification of surface reconstructions on GaAs [11]. There is a strong urgency to accelerate the fundamental research in ML-driven MBE synthesis in the U.S. to reestablish the American competitiveness in advanced precision fabrication.

Future opportunities of putting MBE in the cloud MBE is a deposition technique that fabricates materials layer-by-layer with the highest precision [3, 12] (see Fig. 2). In an ultrahigh vacuum (UHV) chamber with a pressure $< 10^{-9}$ mbar, high-purity molecular beams from thermal effusion cells or gas fluxes such as O_2 and N_2 are pointed toward the heated substrate. The pristine environment and the high-purity molecular beams ensure the quality of MBE-grown materials. *In situ* Reflection High Energy Electron Diffraction (RHEED) allows users to characterize the material qualities with a real-time feedback. This technique has been broadly utilized to fabricate semiconductor III-V materials [13], chalcogenides [5, 14, 15, 16], oxides [3, 17], and nitrides [18]. Moreover, due to the drastically different reaction kinetics at the surface versus the bulk, MBE allows scientists to create artificial superlattices which can only be stable in the thin-film form [17]. The high-quality growth has led to the GaAs/AlGaAs heterostructure with the highest mobility [19], the $FeSe/SrTiO_3$ interfacial superconductor with the highest superconducting transition temperature among all Fe-based materials [14, 16, 20, 21, 22, 23, 24, 25, 26], and the EuO ferromagnetic semiconductor with the highest resistivity contrast across a metal-insulator transition [27, 28].

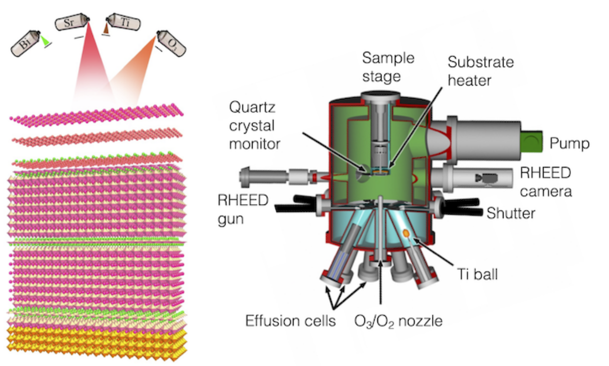


Figure 2: **Illustration of Molecular Beam Epitaxy.** Using dilute, molecular beams in an ultrahigh vacuum chamber, molecular beam epitaxy (MBE) allows materials scientists to create artificial materials layer-by-layer. The illustrations are adapted from Ref. [12].

Despite the many benefits, MBE is known to be a costly and labor intensive tool. In order to optimize the crystal quality, MBE users explore a high-dimensional parameter space defined by the source fluxes, flux ratios, substrate temperatures, as well as pre-growth and post-growth annealing conditions. Moreover, due to the system-to-system difference, a well established recipe in the literature may not be transferrable to different laboratory setups. This leads to a tedious and time-consuming process for materials scientists to explore and optimize the growth recipe for new materials. Anecdotally it can take 3$\sim$5 years to optimize the growth recipe for a tertiary compound such as ABO_3 perovskites. This also suggests a user-to-user difference is frequently seen in the quality of materials, due to different users' levels of experiences. For the academic community, such a time-consuming and tedious process cannot meet the demands of discovering new materials at an ever accelerated pace. For industry, the lack of efficiency and reproducibility in MBE growth is incompatible with the growing manufacturing industry for advanced quantum materials.

Future opportunities in data-driven smart cyber manufacturing ML has emerged as a transformative force in the field of smart cybermanufacturing, revolutionizing the way materials are designed, processed, and optimized. In particular, applications in material science and engineering have experienced significant advancements with the integration of ML techniques. These applications encompass accelerated material discovery and design, process optimization, quality control, and even real-time monitoring and control of complex manufacturing processes. Integrating ML with MBE enables real-time monitoring, control, and optimization, consequently enhancing the growth of complex materials with specific structures and properties [6, 7, 8, 9]. Despite the tremendous potential of ML in smart cybermanufacturing, several opportunities and challenges need to be addressed. One major opportunity lies in the development of more efficient algorithms capable of handling the complexities and uncertainties inherent in material science and engineering, such as multi-scale and multi-physics phenomena, and the integration of domain knowledge into ML models. Additionally, there is a need for robust techniques that can handle noisy, incomplete, or uncertain data, as well as methods for integrating multi-fidelity data sources, and for transferring learned knowledge from one manufacturing process or material system to another. By tackling these opportunities and challenges, ML has the potential to further revolutionize smart cybermanufacturing and unlock new possibilities in material science and engineering.

2 Research Description

Our *grand vision* is to leverage the automated MBE operations and automated real-time RHEED feedback, and to build a robotic system with an artificial intelligence brain to continuously optimize the MBE growth. We term it the *cloud MBE* as a concept for **Future Manufacturing**. For the seed grant period, we will specifically focus on the growth of monolayer FeSe superconductors on $SrTiO_3$ substrates [14, 15, 16].

In analogy to how humans perform complex material synthesis, a robotic system needs **"eyes"** to probe materials, a **"brain"** to learn about the success or failure, and **"hands"** to make a better material. Correspondingly, the main components of a Cloud MBE system include an efficient algorithm to extract features from the characterization data for ML, a deliberate ML model to connect the targeted characterization data with a multi-dimensional input parameter space, and an automated robotic system to execute instructions from the ML model. Given the short period and limited resources of the Future Manufacturing Seeding Grant (FMSG), it is unfeasible to construct and commission a fully automated Cloud MBE. In contrast, we aim to demonstrate the main components of Cloud MBE utilizing existing MBE setups in the Yang Lab and prototypical material systems. In the following we discuss the three Objectives which will allow us to demonstrate the *eyes*, the *brain*, and the *hands* of a Cloud MBE system.

Objective 1: Analyze RHEED Data on FeSe Film and $SrTiO_3$ Substrate for Machine Learning The first milestone is to feed massive real-time characterization data to a ML model. In MBE, RHEED provides convenient real-time characterizations, yet the interpretation of RHEED data often relies on experienced MBE operators and requires multiple years of training. For *Objective 1*, we will use a prototypical system – monolayer FeSe superconductor grown on $SrTiO_3$ substrate [14, 16, 20, 21, 22, 23, 24, 25, 26]. This material is chosen due to its structural simplicity and the strong dependence on the specific growth condition. We will systematically generate RHEED data for different growth conditions and at each step of the growth process. The RHEED images will be decomposed using the Non-negative Matrix Factorization to generate the key patterns, which will be the basis for calculating the deviation from the ideal RHEED image using a designed distance function. This analysis will be key to the supervised learning in *Objective 2*, which aims at modeling the relationship between the ideal RHEED image and the growth condition.

Objective 2: Design Machine Learning Models to Guide Self-Tuning MBE Process First, we will utilize the deep (kernel) learning to model the mapping between growth recipes and the resulting material qualities represented by RHEED images. The resulting model will be embedded in a Bayesian optimization framework, which takes batches of material characterization data with systematically varied control parameters, and makes predictions about the optimal growth recipe. Second, we will employ reinforcement learning algorithms, such as Q-learning or Deep Q-Networks (DQNs), to provide real-time decisions to adjust growth parameters.

Objective 3: Implement a Cloud MBE Prototype for Manufacturing Silver Beam Splitters Ultimately, the ML model will generate instructions for the next iteration of growth process, completing the learning cycle. While it is beyond the funding level of this FMSG to completely revamp the chalcogenide MBE system in the Yang lab to support automated growth, we will demonstrate ML-guided automated growth using a smaller testing system which hosts automated sample movement, source control, sample temperature control, and shutter control. We will modify this existing system to enable ML-guided manufacturing of silver beam splitters. With user-defined splitting ratio at specific wavelengths, this Cloud MBE prototype system will generate testing data by varying the silver flux, deposition time, and substrate temperature, learn from these data, and execute instructions given by the ML model. To our knowledge, this will be the first MBE system which minimizes the human intervention and can open the door to an entire new ground of *cloud synthesis*.

2.1 Objective 1: Analyze RHEED Data on FeSe Film and $SrTiO_3$ Substrate for Machine Learning.

2.1.1 Introduction of the FeSe Growth Process on $SrTiO_3$ Substrate

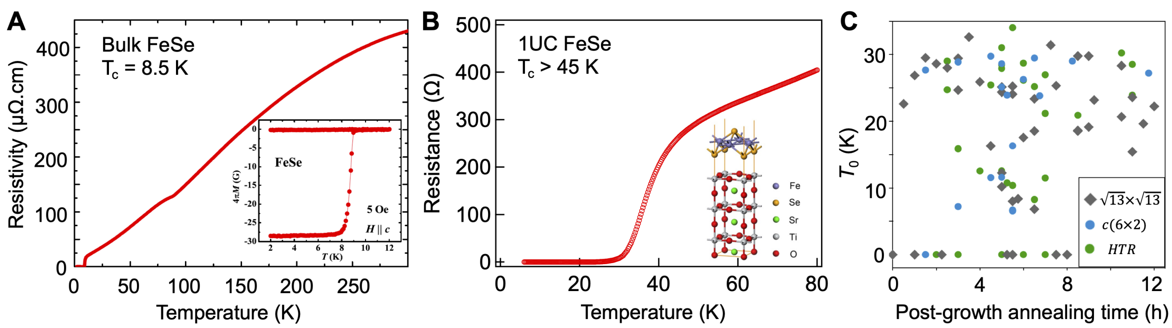


Figure 3: **Interfacial enhancement of superconductivity in 1 UC FeSe/STO.** (A) Resistivity measurement on bulk FeSe characterizing the transition temperature (T_c) close to 8.5 K. The inset illustrates the diamagnetism concomitant with the superconducting transition. (B) Resistance measurement on 1UC FeSe/STO showing a $T_c > 45$ K. (C) Scattered zero-resistance temperature T_0 as a function of the post-growth annealing time. Different symbols represent different STO reconstruction patterns. The T_0 results are scattered with no obvious pattern.

One-unit-cell thick FeSe grown on $SrTiO_3$(100), or 1 UC FeSe/STO, is a unique material platform which exhibits fascinating superconducting properties. Bulk FeSe is an unconventional superconductor with a moderate transition temperature (T_c) of 8.5 K [29, 30], whereas 1UC FeSe/STO exhibits a $T_c > 45$ K according to electrical transport [16], and a $T_c > 70$ K by Scanning Tunneling Spectroscopy (STS) [14] and Angle-Resolved Photoemission Spectroscopy (ARPES) [20, 21, 22, 23, 24]. Here we emphasize that this material serves as an exemplary system which showcases how material qualities can sensitively depend on the specific MBE growth conditions. For instance, Fig. 3 shows that the zero-resistance T_c can scatter in a wide range between 0 and 50 K without obvious correlations with substrate reconstructions or post-growth annealing time. We emphasize that this fact does not suggest that there is no way to control the FeSe growth, but rather demonstrates that the characterization results are complex functions of growth parameters.

We summarize the key parameters which will define a ***recipe***.

Pre-growth substrate annealing. The FeSe growth starts with a delicate treatment of the STO substrate. STO has a pseudo-cubic structure and a strong tendency to form a wide variety of different surface reconstructions [31, 32, 33]. A surface reconstruction such as c(4×2) is characterized by a periodic distortion on the surface layer which has a period four times the lattice constant along one axis and twice along the orthogonal axis. STO can be heat-treated using different conditions to achieve c(2×1), c(4×2), c(4×4), c(6×2), $(\sqrt{13} \times \sqrt{13}$-R33.7°) reconstructions [31, 32, 33]. These reconstructions can be characterized directly by RHEED, which form the data base to train ML models later. Importantly, the FeSe quality sensitively depends on the specific surface reconstruction, which in turn depends on the pre-growth annealing condition defined by the annealing temperature and duration.

Fe and Se fluxes. FeSe grows in the so-called absorption-control mode, for which a Se overpressure is required to grow high-quality FeSe. There is no consensus in the literature on the exact Fe:Se ratio, which ranges from 1:4 [16, 20, 26] to 1:10 [14, 21, 22, 23, 34]. Moreover, the Fe flux determines the growth speed (seconds/layer). There is also no consensus in the literature on the growth speed, ranging from 30 seconds/layer [16, 20, 26] to > 10 minutes/layer [14, 21, 22, 23, 34]. Hence, characterization data will be taken for systematically varied Fe and Se fluxes.

Substrate temperature. The growth temperature ranges between 380 C and 450 C in the literature [14, 16, 20, 21, 22, 23, 24, 26, 34]. It is Yang group's experience that the characterization results depend weakly on the substrate temperature.

Post-growth film annealing. 1UC FeSe/STO as grown is often insulating. To achieve high T_c, the last key parameters for the growth recipe are the temperature and duration for post-growth film annealing. During this annealing, a few important physical processes occur: excess Se will evaporate from the FeSe surface; the crystallinity of FeSe will be further improved; most importantly, the interfacial interaction between the FeSe film and the STO substrate is substantially enhanced. The proposed Cloud MBE system will also systematically learn the dependencies of the film quality on the post-growth annealing condition.

The parameters described above define a *recipe*. **In this FMSG project, the fundamental challenge is to characterize the recipe-quality mapping and train ML models to represent this multi-dimensional map.** Ultimately, the ML model will efficiently converge to the optimal recipe. Thus, it is crucial to define the ***characterization data*** representing sample quality.

Characterization #1: RHEED Images on STO and FeSe. RHEED provides the direct and *in situ* characterization of material qualities. As we will discuss in detail in Section 2.1.2, the quality of 1UC FeSe/STO is best represented by the reconstruction pattern on STO after the pre-growth anneal, and the final RHEED pattern on FeSe after the post-growth anneal. Notably, the RHEED pattern of FeSe reflects the expected symmetry of a 2D square lattice. While it allows the identification of very poor films, it is not a sensitive measure of the superconducting properties. Therefore, it is essential to use the combination of RHEED images from annealed STO and FeSe to define the "quality factor" for supervised ML.

Characterization #2: Transport T_c of 1UC FeSe/STO. The superconducting transition temperature is the most direct measure of the film quality. However, we note that a general MBE setup does not allow a low-temperature *in situ* measurement of electrical resistivity, yet 1UC FeSe/STO is known to be extremely air sensitive. Another approach is to cap the 1UC FeSe film with FeTe and perform *ex situ* transport measurement [15]. This measurement inherently introduces uncertainties in the sample quality, and cannot provide reliable characterization data. We will only perform transport measurements as the final verification of ML-guided recipe. For the future full research grant, we will be able to implement *in situ* transport measurements to obtain reliable T_c's for ML training.

It is worth noting that, PI Yang's previous work [16] demonstrates that ARPES on 1UC FeSe/STO does not give a conclusive characterization of the superconducting property.

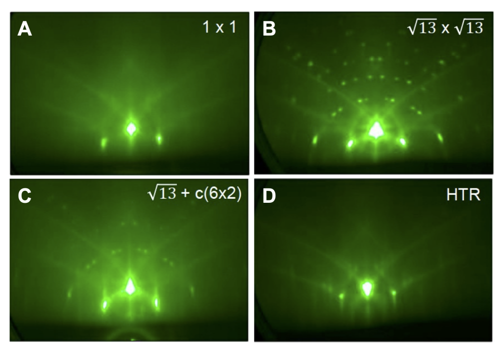


Figure 4: **Examples of RHEED images for STO(100) substrates.** (A) 1×1. (B) ($\sqrt{13} \times \sqrt{13}$-R33.7°). (C) ($\sqrt{13} \times \sqrt{13}$-R33.7°) mixed with c($6 \times 2$). (D) High Temperature Reconstruction (HTR).

2.1.2 RHEED Data Generation and Interpretation

Metric for success: generating $500 \sim 1,000$ RHEED images on STO and FeSe during the 2-year FMSG grant period when varying the growth conditions.

The RHEED image for STO after the pre-growth annealing and the one for FeSe after the post-growth annealing will be used as the main characterization results for supervised ML. **For *Objective 1*, we will obtain these images when systematically varying the MBE growth parameters as defined above, and**

use these data for ML training. For each round of MBE experiments, we will strategically vary only one growth parameter for maximal comparability. In the following we will discuss the expected variety of patterns for STO and for FeSe using preliminary results.

Fig. 4 illustrates different surface reconstructions on STO. While both the $\sqrt{13}\times\sqrt{13}$-R33.7° and c(6×2) reconstructions are stable to vacuum annealing up to the nominal growth temperature of 420 °C, above 600 °C we often observe a transition into a yet-to-be-identified reconstruction pattern which we refer to as the High Temperature Reconstruction (HTR, Fig. 4(D)). Importantly, this HTR pattern has been empirically shown to be correlated with a high chance to obtain high T_c in 1UC FeSe/STO. **Hence, Fig. 4(D) provides a *target* for the Bayesian Optimization discussed later in Section 2.2.**

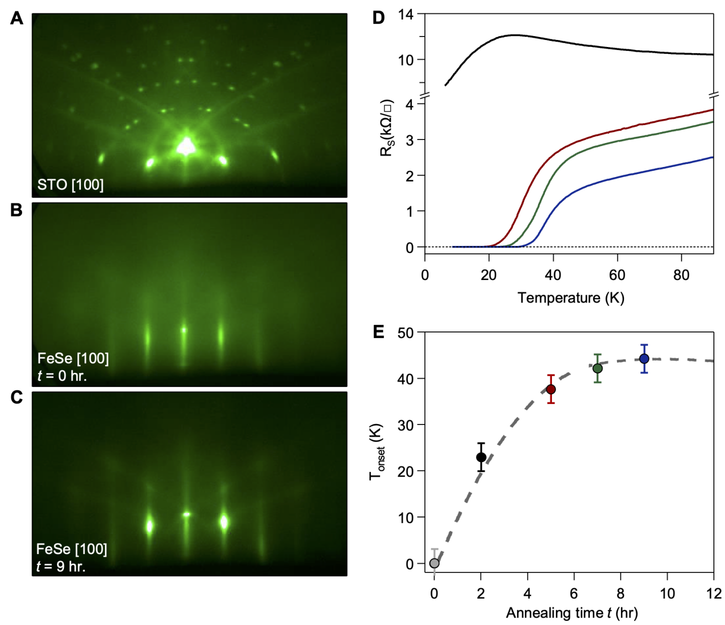


Figure 5: **RHEED images along the [100] direction for** (A) STO substrate prior to film deposition, (B) 1UC FeSe film immeidately after growth, and (C) 1UC FeSe after 9 hours accumulated annealing at 450 °C. (D) Resistivity data as a function of accumulated annealing time. (E) T_{onset} as a function of accumulated annealing time.

Fig. 5 illustrates the typical RHEED evolution for FeSe growth and post-growth annealing. For this particular sample, we obtained the $\sqrt{13}\times\sqrt{13}$-R33.7° reconstruction on STO. The Fe source was set at 1200 °C. The Se source was set at 212 °C. The effective Fe:Se ratio was 1:4. The FeSe growth rate was 40 seconds per layer. Right after the growth of 1UC FeSe, the RHEED pattern completely transformed from the reconstructed STO pattern (Fig. 5(B)). The relevant dots for the $\sqrt{13}\times\sqrt{13}$-R33.7° reconstruction disappeared. This suggests a complete coverage of the FeSe film on the STO substrate, and also demonstrates the surface sensitivity of RHEED. As the last step for growth, post-growth annealing was performed at 450 °C to continuously improve the FeSe film. Upon initial growth, the film exhibited weak RHEED spots and strongly insulating behavior. Upon further annealing at 450 °C, the film eventually became metallic and superconducting, reaching an optimal T_c after 9 hours. It should be noted that the optimal post-growth annealing temperature and duration highly depend on the initial condition of the STO substrate and the as-grown FeSe film. Hence, it is important to use not only the optimal RHEED image for FeSe as the ML metric, but also the optimal RHEED image for STO as an additional metric. The RHEED pattern for the optimally-annealed film shows sharp, well defined spots and distinct Kikuchi lines (Fig. 5(C)), indicating an atomically flat surface with high crystallinity. **For supervised ML models, one should use the RHEED image in Fig. 5(C) as the *target* for the Bayesian Optimization discussed in Section 2.2.**

2.1.3 Non-negative Matrix Factorization on RHEED Data

Metric for success: separate the different patterns in RHEED data which enables the construction of a *distance* function for ML.

In order to simplify the RHEED image analysis, in particular to facilitate an easy design of the *distance* function utilized in Section 2.2, **we will invoke the Non-negative Matrix Factorization (NMF) to decompose the time series of RHEED data into a few representative components.** This analysis allows an easy visualization of the different time evolutions for different features, and thus the identification of the most important features for ML modeling.

NMF is a method to obtain a linear representation of data using non-negative constraints. It enables a low-rank approximation of a complex data set using a few principal components [35, 36, 37]. The basic idea is to represent a 2-dimensional data $\mathbf{V}_{nm}$ by the low-rank, non-negative matrices $\mathbf{W}_{np}$ and $\mathbf{H}_{pm}$, where p is the rank of the approximation.

$$\hat{\mathbf{V}} = \mathbf{WH} \approx \mathbf{V}, \tag{1}$$

If we use n as the spatial dimensions for RHEED and m as the time axis, NMF will effectively decompose the time series of the RHEED data into a few principal components. Each column vector of $\mathbf{W}$ is a *component*. Each row vector of $\mathbf{H}$ represents the corresponding time evolution.

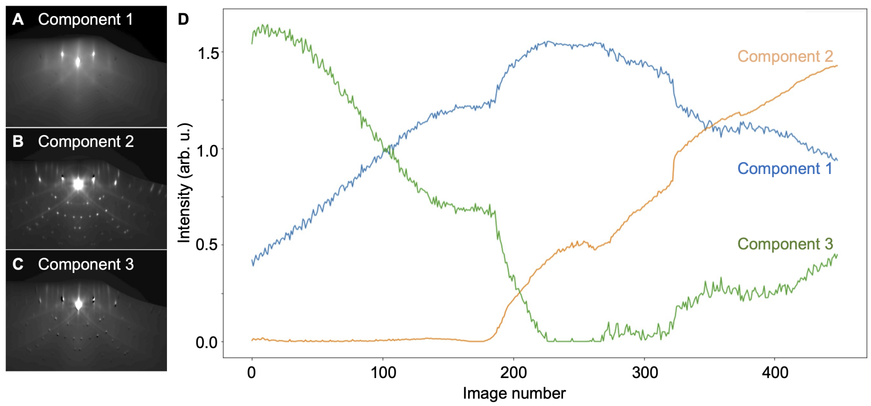


Figure 6: **Non-negative Matrix Factorization (NMF) for a time series of RHEED data on STO** (A-C) RHEED components from the NMF analysis. (D) The time evolution of the 3 RHEED components.

We present a preliminary NMF analysis on the time series of RHEED data during a continuous pre-growth annealing process for STO (Fig. 6). The substrate temperature started around 200 °C, reached 1000 °C at time point #250, and then declined back to 200 °C at time point #450. The rank number of NMF was chosen to be 3 for simplicity. As seen in Fig. 6(A-C), *Component 1* represents a simple 1×1 pattern from the original unreconstructed STO surface; *Component 2* mainly represents the $\sqrt{13} \times \sqrt{13}$-R33.7° reconstruction with a few extra "streaks" above the specular reflection; *Component 3* represents the background intensity between the $\sqrt{13} \times \sqrt{13}$-R33.7° dots. Their respective time evolutions (Fig. 6(D)) provide substantial insight into how the STO surface morphology changes: as the temperature is ramped up, the crystallinity and flatness of the STO surface are improved, leading to an enhanced *Component 1* and a decreased *Component 3*. Around $600 \sim 800$ °C, the $\sqrt{13} \times \sqrt{13}$-R33.7° reconstruction appears. Interestingly, this reconstruction becomes more intense even as we ramp down the temperature. It is worth noting that this preliminary NMF analysis is not meant to represent the typical evolution of RHEED images on STO. In fact, it does not exhibit a pronounced HTR pattern. Nevertheless, it serves as an example to demonstrate how NMF helps the feature extraction and the identification of the crucial features.

To evaluate the deviation of a given RHEED image to the *target* image, it will be more physically meaningful to define the *distance* function using the NMF-decomposed components. A designed weighted sum of the distances when comparing each component can be used to more accurately capture the features relevant for STO annealing and FeSe growth.

2.2 Objective 2: Design Machine Learning Models to Guide Self-Tuning MBE Process

We present a comprehensive approach for developing self-learning MBE models for the growth process of 1UC FeSe/STO. Combining **(deep) Bayesian optimization** and **reinforcement learning**, we aim to leverage the massive collection of experimental data for systematic planning and real-time control, optimizing the growth process and achieving improved material quality.

2.2.1 Data-driven MBE Modeling

To effectively handle the complex relationships between the input features and desired material properties, we propose a deep learning model that can process the data described in the previous section, which includes continuous experimental growth parameters such as temperature, deposition rate, substrate orientation, and annealing time, etc., as well as RHEED image-derived metrics. These metrics could be in the form of a quantitative score representing the distance to a target response pattern or a more accurate measurement of the output, such as the material resistance, which is more expensive to obtain but more directly reflects the target to be optimized.

Model architecture. A suitable choice for this task is a Convolutional Neural Network (CNN) combined with a fully connected layer. The CNN is capable of handling the spatial relationships within RHEED images and extracting meaningful features. These features can then be used to compute the RHEED image-derived metrics. To process the continuous experimental growth parameters, we can use a fully connected (dense) layer. This layer will be concatenated with the output from the CNN after flattening it, which enables the model to learn interactions between the features extracted from the RHEED images and the continuous experimental growth parameters. Finally, the concatenated output will be passed through a series of fully connected layers with appropriate activation functions, leading to the final output layer. This output layer will provide the desired target, such as material resistance or distance to a target response pattern.

Uncertainty quantification Incorporating predictive uncertainty in our deep learning model can help assess the reliability of predictions and enable more informed decision-making during the MBE process. To capture predictive uncertainty, we can employ deep ensembles [38] or a deep kernel learning [39, 40, 41] framework. Deep ensembles involve training multiple models with different initializations and then combining their predictions. This approach is conceptually straightforward and has been shown to provide a robust approximation of model uncertainty. For our task, **we will train several instances of the proposed neural network** (CNN and fully connected layers with different random initializations), and then **use predictive variances across the ensemble members to estimate the model's predictive uncertainty**. Alternatively, we can use a deep kernel learning framework to capture predictive uncertainty. In this framework, **we will use the proposed neural network as feature extractors, followed by a Gaussian process layer to model the relationship between the features and the output**. Given a set of training examples S consisting of experiment recipes x_S and the system response y_S, one can jointly learn the neural network weights $\mathbf{W}$ and hyperparameters θ of the kernel by differentiably maximizing the marginal log-likelihood function given observations:

$$\log(y_S | x_S, \mathbf{W}, \theta) \propto -[y_S^T K_S y_S + \log |K_S|],$$

Here, K_S is the covariance matrix induced by the kernel function which can be used to provide a probabilistic prediction of the output.

Both deep ensembles and deep kernel learning frameworks can provide valuable information about predictive uncertainty in our deep learning model for MBE optimization. By incorporating these uncertainty estimates into the decision-making process, we can enhance the robustness and efficiency of the MBE growth process and achieve better material quality.

2.2.2 Adaptive Recipe Design for MBE via Deep Bayesian Optimization

Metric for success: predicting the optimal and repeatable recipe for FeSe growth on STO using Deep Bayesian Optimization.

Bayesian optimization in MBE Bayesian optimization (BO) is a popular approach for optimizing expensive black-box functions and has been successfully applied in various domains, including material science and molecular engineering. In this project, **we will employ Gaussian processes with deep kernels or deep ensembles in a Bayesian optimization framework to optimize the MBE growth process**. For proposing new experiments, we can employ standard acquisition functions such as upper confidence bound [42] or a custom acquisition function tailored to the specifics of the MBE growth process. The acquisition function quantifies the utility of sampling a particular point in the input space, guiding the exploration-exploitation trade-off in the Bayesian optimization process.

Handling multi-fidelity responses In the MBE growth process, we encounter two types of responses that represent different levels of fidelity. On the one hand, high-fidelity responses, such as material resistance, provide an accurate measurement of material quality but are expensive to obtain. Acquiring this data requires removing the sample from the chamber after a specified annealing duration and physically measuring the resistance, which renders the sample unusable. Moreover, the air sensitivity of the sample makes the resistivity measurement unreliable, as discussed in Section 2.1.1. On the other hand, low-fidelity responses, such as the distance to a target RHEED pattern, can be monitored in real-time and serve as a more affordable proxy for the material quality.

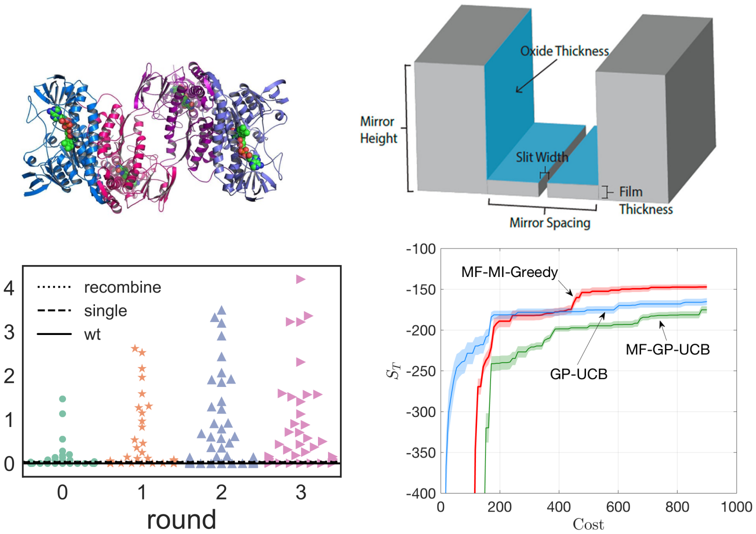


Figure 7: Prior work by **Co-PI Chen** on **Bayesian optimization for optimal experimental design** (Left) Batched Protein design on the GB1 dataset [43] (Right) Mirrored plasmonic filter design [44]. The bottom plots show optimization trial results for each application. The x-axis reflects experimental cost, and the y-axis represents experimental utility (higher values indicate better performance).

We will employ Multi-Fidelity Bayesian Optimization (MFBO) to reduce reliance on costly high-fidelity measurements while still capturing underlying relationships. MFBO models high-fidelity and low-fidelity responses using separate Gaussian processes, with the latter serving as a correction term for the former. This allows the model to capture differences between fidelity levels and weigh their contributions to the optimization process. As new data points are acquired, the model updates its predictions, refining the surrogate function and improving optimization results. As demonstrated in Fig. 7, **Chen** has extensive experience in MFBO applied to related tasks in physical sciences, including nanophotonics and biochemical engineering. **We will adapt these techniques to the MBE growth process, to efficiently optimize experimental parameters while minimizing the need for expensive high-fidelity measurements**, ultimately leading to better material quality. Furthermore, to enhance the efficiency of the BO process, **we will leverage the existing offline data to pretrain the model**, such as the neural network weights in the deep kernel or the deep ensemble members. This can warm-start the experimental design, reducing the need for costly online exploration.

2.2.3 Real-time Optimal Control for the MBE Growth Process

Metric for success: demonstrating a preliminary real-time instruction for guiding the FeSe growth.

The end goal of our research is to develop a system that can control the MBE growth process in real-time, similar to how a human operator constantly monitors the RHEED image and adjusts growth parameters accordingly. In order to enable real-time control and adjustment of the current recipe, we have to consider "state-dependent action sets" (e.g., incremental adjustments to an existing recipe), where vanilla BO approaches investigated in the previous task (which recommend "full recipe" from scratch) no longer apply.

Reinforcement learning and optimal control Naturally, we formulate the MBE growth process as a Markov Decision Process (MDP) problem which can be solved via reinforcement learning (RL) algorithms. In the context of MBE real-time control, an MDP process considers an agent taking actions such as adjusting the deposition rate, substrate temperature, and other growth parameters in response to the current environment state (e.g., RHEED image patterns, growth stage). The agent receives a reward based on the distance to the target pattern or whether the target pattern is achieved and then transits to the next state. The reinforcement learning algorithm learns a policy to interact with the environment to optimize the cumulative reward received at the end of the MBE process.

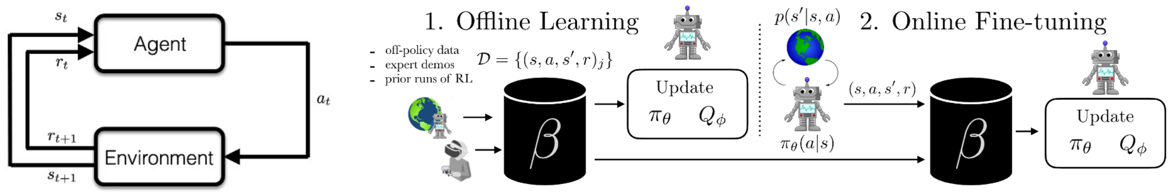


Figure 8: **Offline reinforcement learning with online fine-tuning** for real-time optimal MBE control. (Left) An MDP diagram showing the RL algorithm ("agent") interacting with the MBE environment through sequential actions a_t at state s_t. (Right) The proposed offline-to-online RL framework uses offline RL to warm start the online learning process. Here, β represents offline or online RL algorithms, and π_θ, Q_ϕ are the learned policy and reward function.

To improve online sample efficiency, we propose using offline reinforcement learning with online fine-tuning (as illustrated in Fig. 8). Conventional RL requires costly and time-consuming online interactions. However, if we have already collected data on a similar environment or task, such as the time series demonstrated in Section 2.1.3, we can use it as an offline dataset to pre-train our RL algorithm. Offline reinforcement learning enables training RL policies purely from pre-collected datasets without any online exploration, translating massive offline datasets into powerful decision engines [45]. This approach is particularly impactful in real-world applications like robotics and optimal control [46, 47], where online interactions can be expensive. **Chen** is experienced in developing offline reinforcement learning algorithms for optimal control. We will build on a suite of offline reinforcement learning algorithms, including Conservative Q-Learning (CQL) [48], Batch-Constrained Q-Learning (BCQ) [49], Offline Reinforcement Learning with Implicit Q-Learning (IQL) [50], Uncertainty-Based Offline Reinforcement Learning with Diversified Q-Ensemble (EDAC) [51], and combine them with online fine-tuning techniques for real-time control [52, 53]. The success of these algorithms in various robotic control tasks inspires us to generalize their application to the MBE growth process, effectively leveraging pre-collected datasets while enabling real-time adjustments to the growth parameters.

Leveraging human feedback at uncertain states Incorporating human assistance in the learning process, particularly when it comes to addressing novel and unexpected system responses, can significantly improve the overall performance and reliability of the RL algorithm. The proposed *distance to a target pattern* is not always a gold-standard reward function—when encountering abnormal or unseen states, the RL algorithm should summarize these states and ask experts (material scientists) for feedback (e.g., whether the observed pattern is better than the target pattern). By incorporating human feedback, we can better align the RL algorithm's decision-making process with expert knowledge, leading to a more robust and efficient control system. To effectively incorporate human feedback in the MBE growth process, **we propose to use a variant of the Proximal Policy Optimization algorithm [54], such as PPO-HF [55], to probe domain experts for additional feedback in the learning process**. PPO-HF operates by training a policy using a batch of collected data, which includes states, actions, and rewards, along with any human feedback on those actions. Human feedback is integrated as an additional reward signal, which guides the policy update. When the RL agent encounters an abnormal or unseen state during the MBE growth process, it can request human feedback on the proposed action or a set of alternative actions. The human expert can provide feedback on the quality of the proposed actions, and this information is used to update the policy.

2.3 Objective 3: Implement a Cloud MBE Prototype for Manufacturing Silver Beam Splitters

Metric for success: demonstrating the automatic ML-informed synthesis of silver beam splitters according to user specifications.

In Section 2.1 and 2.2, we have discussed the implementation of the **"eyes"** and the **"brain"** of a robotic Cloud MBE system. The third crucial component in a fully automatic system is to implement the **"hands"** which execute instructions from the ML models and manipulate various hardware control knobs to complete the synthesis. Meanwhile it is beyond the budget and scope of this FMSG to convert our main FeSe MBE system into a fully automatic system. **Hence, we aim to demonstrate the principles of ML-guided automatic synthesis using a small-scale testing MBE system also hosted in the Yang lab at the University of Chicago.** We define the growth task as manufacturing silver thin films as optical beam splitters with user-specified splitting ratio and operational wavelength. Compared to the fabrication of FeSe, this growth task involves only one effusion cell and is less sensitive to the pre-growth annealing. The growth *recipe* can be defined by source temperature, growth time, substrate temperature, post-growth annealing temperature, and post-growth annealing time. Instead of using RHEED data as the metric of the film quality, we will use *in situ* reflectivities and transmissivities measured at a few discrete wavelengths.

2.3.1 Existing Prototype Hardware: Automated Sample Transfer and Real-Time Data Output

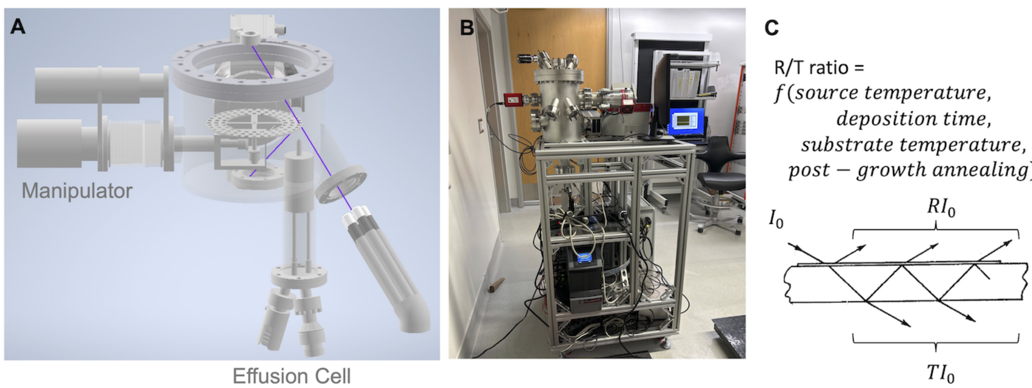


Figure 9: **A Cloud MBE prototype for synthesizing silver-film beam splitters.** (A) The design concept of the prototype system consisting of a motorized manipulator which can deposit 90 beam splitters, one silver effusion cell, and an *in situ* optical reflectivity and transmissivity setup using 5 laser pointers at different wavelengths. (B) The setup in the Yang lab. (C) The concept of controlling the source temperature, deposition time, substrate temperature, and post-growth annealing conditions to achieve specific splitting ratios at user-defined wavelengths..

We will utilize an existing cloud MBE prototype system in the Yang lab for this task (Fig. 9). Importantly, all the sample manipulation, effusion cell control, and characterization data collection have been integrated into a centralized, home-made mission-control user interface which is fully customizable. Our innovative design of a carousel-based sample manipulator puts individual substrates in a circular pattern, and utilizes the rotational and translational degrees of freedom to automatically move from one sample to the next. Real-time characterization data will be taken using the laser reflection and transmission setup, which will output reflectivity and transmissivity of each fabricated beam splitter at > 5 wavelengths. This setup is thus capable of completing the sample transfer, effusion cell control, deposition control, characterization data collection completely without human intervention. Notably, for each round of deposition we will fabricate a total of 90 samples with systematically varied growth recipes, generating > 450 data points for the reflectivities and transmissivities at different wavelengths. At the current stage, all the hardware control has been commissioned and tested, awaiting the integration with the **"brain"** of the Cloud MBE system.

Even though this is a different deposition task, we emphasize that the ML framework we will have established in the previous two sections will be directly transferrable. Here we define the key parameters for a *recipe*, and the nature of the *characterization data*.

Recipe. The recipe of fabricating a silver-film beam splitter is defined by the source temperature, deposition time, substrate temperature, and post-growth annealing conditions. The source (silver) temperature will be varied between $700 \sim 900$ °C, which corresponds to the typical vapor pressure of $10^{-5} \sim 10^{-3}$

mbar. The deposition time will be varied between 0 and 10 minutes, considering the typical deposition rate of 1 nm/min. For high-quality metal-film deposition [56], the substrate temperature will be varied between the liquid N_2 temperature (77 K) and room temperature. A miniaturized cryostat will be purchased to enable the substrate temperature control. The post-growth annealing temperature and duration will be in the range of [20, 200] °C and [0,20] minutes, respectively.

Characterization data. **The goal of the beam splitter fabrication is to realize a user-defined beam splitting ratio at a desired wavelength.** The beam splitting ratio is a complex function of the film thickness, flatness, morphology, and other parameters. Practically, since the splitting ratio is always a smooth function of the laser wavelength (Fig. 10(A)), we plan to take real-time reflectivities and transmissivities at a few discrete wavelengths (e.g. 405 nm, 525 nm, 640 nm, 685 nm, 830 nm). A direct polynomial interpolation of the reflectivities and transmissivities at the user-specified wavelength will be an accurate estimate of the actual optical properties at that wavelength. The *distance* function can be defined either using the measured optical properties at the discrete wavelengths, or using the interpolated values at the desired wavelength. Either way the ML model as defined in Section 2.2 will allow a rapid convergence of the growth parameters.

2.3.2 Implement the Real-Time AI Brain in the Cloud MBE Prototype

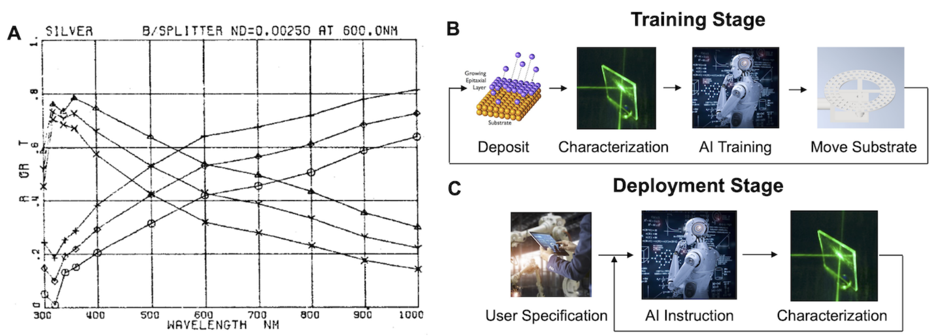


Figure 10: **The operation cycles of a fully automatic cloud MBE prototype.** (A) Reflectivities and transmissivities for silver-film beam splitters, from Ref. [57]. The film thickness is 1.5 nm. The circles, plus signs, and diamonds represent the reflectivities for the p, s, and mixed polarizations. The triangles, crosses, and "Y" signs represent the transmissivities for the p, s, and mixed polarizations. (B) The operation cycle during the training stage. (C) The operation cycle during the deployment stage.

With the defined *recipe* and *characterization data*, we will be able to utilize deep kernel learning and deep ensembles to devise an ML model which serves as the *brain* of the Cloud MBE prototype system. Notably, this prototype system will be run in two stages. At the **Training Stage**, the robotic system generates large volumes of data by systematically varying the growth recipe and collecting the characterization data. The purpose is to train the CNN model to be used for the second stage. The MFBO algorithm developed in Section 2.2.2 will enable us to efficiently optimize the growth recipe for any given wavelength. Notably, even though theoretical electrodynamics allows one to calculate the reflection and transmission of silver films, in realistic devices the smoothness and grain sizes of the silver films will also substantially alter the optical properties. Hence, the growth recipe is not *a priori* known and a ML model will greatly enhance the efficiency and accuracy for converging to the optimal efficiency. At the **Deployment Stage**, a user will specify the beam splitting ratio at a particular wavelength. The ML model takes this user input and generates instructions to fabricate the desired beam splitter. It is expected that a few iterations between AI instructions and beam splitter fabrication will lead to the optimized manufacturing. Moreover, the RL algorithmm developed in Section 2.2.3 will enable the Cloud MBE prototype system to adjust its growth conditions in real time. As such, we will demonstrate a completely new paradigm for **Future Manufacturing**: the machine self-learns the material recipe and deploys this recipe to automatically make a product according to the user specification. The human element is removed.

Finally, we remark that even though the **"eyes"**, the **"brain"**, and the **"hands"** of a Cloud MBE system is demonstrated through different testing systems available in the Yang lab, these proof-of-principle experiments will allow us to generalize the framework to significantly more complex MBE systems and material systems. Moreover, the same philosophy of moving synthesis sytems to the cloud can broadly apply to soft materials synthesis as well [2], which will altogether lead to a bigger scope for establishing a research center for *Cloud Synthesis*. Therefore, the success of the current FMSG grant will lead to an exciting Future Manufacturing full proposal.

2.4 Teaming Plan

2.4.1 Management of the Current Team

PI Yang is a materials science expert who specializes in the growth of chalcogenide thin films. **Yang** has demonstrated success in fabricating FeSe/STO thin films using traditional human-driven MBE [16, 25, 26, 58], and will be responsible for the hardware implementation of the Cloud MBE project. **Yang**'s group will utilize the existing FeSe MBE and the testing Cloud MBE systems to perform systematic synthesis of FeSe/STO and ultrathin silver films, feeding large volumes of data to the ML models. **Co-PI Chen** is a computer science expert who leads various large-scale collaborations on AI-in-Science within and beyond the UChicago campus. **Chen**'s previous work has demonstrated that ML can successfully optimize the design parameters for various scientific applications, including a plasmonic filter design [44, 59], cosmic experimental design [40], and biochemical engineering [43]. The **Chen** group will be responsible for analyzing the data from the **Yang** group, and design ML and RL models to predict the optimal growth recipe. The two groups will coordinate through monthly team meetings and frequent interactions via informal meetings. **Yang** will keep track of the overall project timeline and ensure that the team anticipates and addresses bottlenecks proactively.

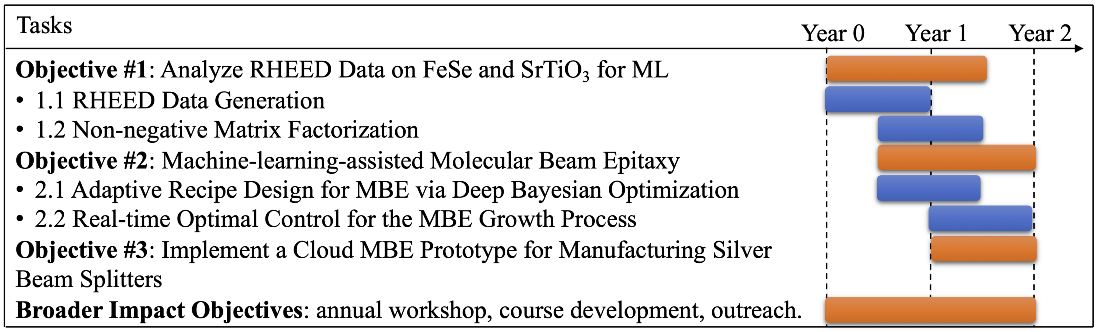
2.4.2 Team Expansion and Industry Collaboration

In anticipation of expanding the project into a Future Manufacturing full proposal, the team will actively seek opportunities to partner with materials scientists, physicists, engineers, and computer scientists at the University of Chicago. We have already started this effort through the conversation with Prof. Supratik Guha at the Pritzker School of Molecular Engineering and Dr. Jie Xu at Argonne National Laboratory (letters of collaboration appended). We will further crystallize the intellectual conversations on Cloud Synthesis by organizing an annual *Cloud Talks* workshop inviting interested faculty members, national lab scientists, industrial leaders, as well as postdoctoral scholars and graduate students to a half-day intellectual exchange forum. It is one of our most important Broader Impacts objectives to seed and grow this intellectual conversation, which will eventually lead to a group of 5$\sim$6 principal investigators to form the basis for a Cloud Synthesis hub in the Midwest region.

2.5 Risk Assessment and Mitigation

Leveraging **PI Yang**'s substantial experience in the FeSe/STO fabrication [16, 25, 26, 58], the risk for collecting systematic RHEED data from 1UC FeSe/STO films is low. Nonetheless, due to the intrinsic uncertainties in the substrate quality, the optimal growth recipe may exhibit a batch-to-batch variance. To mitigate this risk, we will closely coordinate with the substrate vendor (Shinkosha, Co. Ltd.) to obtain a large number of consistent substrates from the same batch. For *Objective 3*, it is important to note that the laser power during a long deposition cycle can drift. To mitigate this risk, we will split a small fraction of the laser power as a calibration source, and normalize the instant readings according to the calibration laser power.

2.6 Project Timeline



3 Results of Prior NSF Support

PI: Shuolong Yang is supported by *NSF DMR-2145373*; Amount $687,153; 02/2022-01/2027; **"CAREER: Tuning Topology and Strong Correlations for the Next Generation of Topological Superconductors".** *Intellectual Merit:* This project will provide a complete synthesis and physics understanding of the novel topological superconducting thin films FeTexSe1-x on oxide substrates. We will provide answers to a number of pressing issues in topological physics, such as (1) the nature of the topological phase transition, (2) the mechanism of correlation-enhanced superconducting order, and (3) how to realize nonabelian anyon statistics. *Broader Impacts:* The knowledge and expertise developed in this project will not only be groundbreaking in the development of topological quantum computing, but will also lead to educational and outreach activities under the theme of *"Immersive Quantum Material Education."* **Publication:** [60]

Co-PI Chen is currently supported by three NSF grants, and the most related is *NSF-2037026*; Support period: 01/01/2021-12/31/2025; Total amount: $10,000,000 (in total); Chen's share: $375,000; **"FMRG: Manufacturing ADvanced Electronics through Printing Using Bio-based and Locally Identified Chemicals"**. *Intellectual merit:* Develop novel machine learning algorithms for optimizing the design of the manufacturing supply chain from precision agriculture/hydroponics to advanced biodegradable and recyclable electronics. *Broader impacts:* An open-sourced democratized manufacturing prototype that enables distributed manufacturing of low-cost printable electronic devices using locally identifiable resources such as bio-based materials derived from plants. Educate and train a wide audience on the proposed future manufacturing using evidence-based practices grounded in experiential learning. **Publication:** [41, 61].

4 Broader Impacts

This project will potentially transform how to perform high-precision MBE deposition in the future, and build a direct interface with manufacturing in industry. In particular, the technology of Cloud MBE will lead to direct applications in optical devices, superconducting devices, and quantum information devices. We will work closely with the Polsky Center for Entrepreneurship and Innovation at the University of Chicago to properly disclose the technology through appropriate channels. We are aware of the disconnect between academic and industrial sectors on the MBE activities, and will host an annual *Cloud Talks* to invite academic and industrial leaders as a forum for not only our team members, but also the extended collaborative network and industrial partners to exchange ideas. We will also leverage the geographical locations of South Side Chicago, which is immersed in the African American community, and provide outreach opportunities to engage the under-represented minority students. This project will enable us to educate, train, and connect the next-generation workforce for materials synthesis, which is key to reviving the American manufacturing industry in the Midwest region.

4.1 Education and Workforce Development

Graduate and Undergraduate Student Mentoring The PI and Co-PI will train the new generation of workforce by closely co-mentoring graduate and undergraduate students. It is worth noting that PI **Yang** has been mentoring a team of 3 undergraduate students to build the first Cloud MBE prototype system as mentioned in Section 2.3. One of the students is a female student. **Yang**'s group is also the only quantum materials group at the University of Chicago who has successfully recruited an African American PhD student. The students working on this project will participate in the annual *Cloud Talks* workshop to directly interact with industrial leaders, which will facilitate potential matching between the students and future job opportunities in the Midwest region.

Course Development The PIs see great importance in developing courses tightly integrated with research. At UChicago, Co-PI Chen has already developed a new graduate course on *Bayesian Optimization and Adaptive Experimental Design*, which covers several aspects related to this proposal. We will integrate the methods investigated in this project into this course. PI Yang will also develop a new course "Science of Materials" which aims at starting with the core solid-state principles and going beyond the standard single-particle description at the end of the course. This course will be part of the new materials science curriculum under development within the Pritzker School of Molecular Engineering, and prepare students for more advanced quantum engineering and quantum materials studies. PI Yang will dedicate 30% of the new course content to frontier research topics such as advanced manufacturing, including the new paradigm of *Cloud MBE* manufacturing.

4.2 Annual Workshop: Cloud Talks

We will host an annual *Cloud Talks* workshop to invite the FMSG team members, extended collaborators, and industrial partners to join in a half-day workshop, which aims to provide a forum for the team to have close interactions and outline immediate and long-term goals for future Cloud Synthesis activities. Moreover, the workshop will help broaden the impact of our FMSG project to regional materials science communities. The workshop will feature speakers from outside the team and will provide annual summaries of updates in the broad field of *smart manufacturing*. In particular, we will join force with the existing effort of AI-driven robotic synthesis of polymer materials at Argonne National Laboratory, and crystallize ideas and concepts for a regional *Cloud Synthesis* center for both hard and soft materials. The University of Chicago has a strong interface with the manufacturing industry in the Midwest region (e.g. Veeco, 3M, Seagate).

4.3 Outreach Activity: UChicago Quantum Quickstart

The UChicago Quantum Quickstart program is a 1-week accelerated quantum introductory workshop for students from the 9th to 11th grade. Yang is one of the two inaugural instructors for the 2021 program, which received 276 total applications with 11% identified as African-Americans, 22% as Hispanics/Latinos, and 1% as Native Americans. A total of $\sim$70 students were recruited. Under this proposal, PI Yang will leverage the existing UChicago Quantum Quickstart program and incorporate frontier research results from *Cloud MBE*. This has a particularly strong impact due to the strong presence of manufacturing in America's Midwest region, and will encourage under-represented students to envision and pursue careers related to manufacturing from an early stage.

4.4 Outreach Activity: Physics with a Bang

The Yang group has been active participants of the "Physics with a Bang" outreach program hosted by the University of Chicago community involving more than 500 participants from K-12 students and local residents. We have a traditional program to demonstrate quantum levitation using high temperature superconductors. Such activities promote awareness among the general public on the amazing properties of quantum materials and how they may allow future technologies to be developed. Under this project, we will utilize the compact testing Cloud MBE system, and make a demo booth at the Physics with a Bang event. Importantly, the testing system is a fully mobile system with an uninterrupted power supply providing 6 hours of stand-alone operation. We will be able to engage with the public and let them specify the parameters for optical beam splitters and start the automatic "cloud" synthesis. This will be an opportunity to showcase the amazing future promises of Cloud MBE by directly engaging the general public.